JFEM Shell Element Formulation, Materials, and Internal Gates

A code-driven companion to the JFEM architecture manuscript

JFEM

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Abstract

This document is a focused, code-driven description of the shell-element machinery currently implemented in the JFEM Julia finite-element solver (implementation snapshot described below). It is a companion to the previous architectural manuscript shipped under `Reference_documentation/jfem_2026_architecture_manuscript.pdf`, which covered the parse/solve/export workflow, the JSON interfaces, the supported `SOL` families, the adjoint infrastructure and the broad validation harness. Rather than repeat that material, this note expands the parts of the prior paper that were only summarised: the kinematics and integration rules of every shell kernel actually present in `src/FEMKernels.jl`, the constitutive paths for `MAT1`, `MAT2`, `MAT8` and `PCOMP`, the assembly-level dispatcher in `src/solver/assembly.jl` that selects a kernel for each `CQUAD4` and `CTRIA3`, and the complete catalogue of environment variables and `PARAM` switches (the *internal gates*) that change shell behaviour. The geometric-stiffness and stress-recovery pieces that shells share with the buckling and adjoint paths are documented at the same level. Every claim has a file:line pointer to the corresponding source. The intent is precision: the reader should be able to predict the operator assembled for a given deck and `ENV` configuration without running the solver.

Keywords: finite-element shells; MITC4; MITC4+; DKQ; DKMQ; Adini plate; MIN4; MacNeal residual-bending flexibility; Hu–Washizu hybrid stress; classical laminate theory; PCOMP; PSHELL; geometric stiffness; internal gates; Reference parity.

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1 Introduction and scope

JFEM is a compact Julia finite-element solver with a BDF-compatible front end. Its overall architecture — the three-stage parse/solve/export workflow, the JSON-serialisable internal model, the supported solution sequences (`SOL 101`, `103`, `105`, experimental `106`), and the adjoint back-end — is documented in the previous manuscript shipped alongside this one. That manuscript stays the right starting point for new readers.

What it does *not* document in depth is the shell layer. The previous paper deliberately gave a short, paragraph-level treatment under "Discretisation choices" and noted that the shell code in `src/FEMKernels.jl` and `src/solver/assembly.jl` contains "selective reduced integration, incompatible-mode toggles, curved-shell normal handling, drill-stiffness controls, and multiple heuristics specific to buckling accuracy". That single sentence covers, in practice, several thousand lines of code, around a dozen distinct kernel formulations, four separate material pathways, and roughly seventy environment-variable gates.

This document is the long version. It is restricted to shells: rod, bar, beam, spring, rigid, mass and solid elements are out of scope. The solution-sequence dispatch and the export layer are also out of scope; the shell sections of the prior paper that summarised them are treated as background.

The document is organised as follows. Section 2 restates the small amount of architectural context needed to read the shell discussion. Section 3 fixes the kinematic conventions and DOF layout. Section 4 walks through every shell material pathway (`MAT1`, `MAT2`, `MAT8`, `PCOMP` CLT, `MATT1` thermal continuation, `PSHELL` bookkeeping). Section 5 documents the `CQUAD4` dispatcher with its full decision tree. Section 6 gives, for each `CQUAD4` kernel, the formulation, integration rule, options and literature reference. Section 7 covers the `CTRIA3` macro-quad triangle. Section 8 covers the subdivision of `CTRIA6` and `CQUAD8`. Section 9 documents the shell geometric-stiffness operator and the stress-field classifier used in `SOL 105`. Section 10 covers shell stress recovery. Section 11 covers consistent mass. Section 12 lists every gate exposed by the shell layer (environment variables, `PARAM` entries and global `Ref`s). Section 14 closes with the current limitations and the open parity items.

2 Architecture recap relevant to shells

JFEM is organised around three stages: *parse* (`BDF` → internal model), *solve* (assembly and one of `SOL 101/103/105/106`), and *export* (VTK, JSON, HDF5, native binary). The shell layer is exercised by all four solution sequences. The key facts the shell discussion relies on are:

- All shells carry six DOFs per node: three translations and three rotations. The drilling DOF is always present at assembly time, but the solver distinguishes between an *active* drill (penalty stiffness from `PARAM,K6ROT`) and a fully released drill (membrane-only properties, see §4).
- Shell properties (`PSHELL`, `PCOMP`) are normalised by the model builder into a unified *shell-property dictionary*: keys `Cm`, `Cb`, `Cs`, `Bmb` (membrane, bending, shear, membrane–bending coupling), plus per-ply metadata for composites. Element kernels consume this dictionary directly and do not see the original card form.
- Higher-order shell cards (`CTRIA6`, `CQUAD8`) are not served by dedicated quadratic kernels. They are decomposed during assembly into sub-elements that go through the same `CTRIA3` and `CQUAD4` pipelines (§8).
- Every shell kernel is dispatched per element, in a single pass, by `src/solver/assembly.jl`. The dispatcher reads the property type, the geometry, the active solution sequence, a set of environment variables, and a set of `PARAM` controls, and chooses one of the kernels documented in §6–7.

- The `SOL 105` buckling and `SOL 103` modal paths use the same direct/Krylov dispatch logic as Reference [15]; the shell layer sees the same operator regardless of which eigensolver is invoked.

3 Kinematics, frames and the shell DOF layout

Reissner–Mindlin kinematics. All checked-in shells are based on first-order shear-deformable kinematics [1, 3]:

$$\mathbf{u}(x, y, z) \;=\; \mathbf{u}_0(x, y) \;+\; z\,\boldsymbol{\theta}(x, y), \qquad |z| \le h/2, \tag{1}$$

with $\mathbf{u}_0 = (u, v, w)$ the mid-surface displacement, $\boldsymbol{\theta} = (\theta_x, \theta_y)$ the rotations of the cross-section (no thickness stretching), and h the shell thickness. Transverse shear is retained in the standard form $\gamma_{xz} = \partial w/\partial x - \theta_y$, $\gamma_{yz} = \partial w/\partial y + \theta_x$, modified at the element level by the locking-prevention mechanisms documented in §6.

DOF ordering. Each shell node carries the six standard Reference DOFs in the order $(u, v, w, \theta_x, \theta_y, \theta_z)$. The drilling component θ_z is assigned a penalty stiffness controlled by `PARAM,K6ROT` (§12), built in the Hughes–Brezzi form [6] that couples in-plane rotations to the membrane gradient, with the geometrically-exact and computational treatments of Ibrahimbegović and Frey [17, 18] providing the underlying theoretical framework for drilling-aware shells. The local element stiffness is 24×24 for `CQUAD4` and 18×18 for `CTRIA3`; both are transformed to the global frame before assembly.

Element frame. Every shell carries a local frame $(\mathbf{t}_1, \mathbf{t}_2, \mathbf{n})$. For `CQUAD4` this is computed from the corner coordinates; for warped quads the geometric normal is the average of the two diagonal cross-products and the in-plane axes are orthonormalised against $\mathbf{n}$. The frame can be overridden by the `THETA`/`MCID` field of the element card; for `PCOMP` the per-element material rotation is applied to $\mathbf{C}_m$, $\mathbf{C}_b$, $\mathbf{C}_s$ and $\mathbf{B}_{mb}$ before any other gating decision (see code at `src/solver/assembly.jl` around the `shell_pcomp_material_rotation` helper).

Normals (`SNORM`). JFEM honours `PARAM,SNORM` nodal-normal smoothing. The smoothed nodal normals are used to build a *geomnormal frame* on the curved-shell branches (the `JFEM_Q4_FLAT_CURVED_ISO_GEOMNORMAL_FRAME` and `JFEM_Q4_CURVED_ISO_GEOMNORMAL_FRAME` families, §12); when those gates are inactive the element keeps its local frame from the corner geometry only.

4 Material handling for shells

This section documents how every supported material card reaches the shell kernel. The implementation is split between `src/parsing/extract_properties.jl`, `src/parsing/extract_materials.jl` and `src/ModelBuilder.jl`. The end state, for every shell element, is the 4-tuple $(\mathbf{C}_m, \mathbf{C}_b, \mathbf{C}_s, \mathbf{B}_{mb})$ plus the shell thickness h, the reference modulus E_{ref} used by the drill term, and the per-ply data needed for composite stress recovery.

4.1 Isotropic `MAT1`

The default isotropic path is the simplest one. For `PSHELL/MAT1` with constants (E, ν, h) the kernel builds the constitutive blocks in the classical plane-stress form (verified at `src/FEMKernels.jl:1327-1335`):

$$\mathbf{C}_m = \frac{E\,h}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}, \quad \mathbf{C}_b = \frac{r_b\,E\,h^3}{12(1-\nu^2)}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}, \tag{2}$$

$$\mathbf{C}_s = k_s\,G\,h\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad G = \frac{E}{2(1+\nu)}, \quad k_s = (\mathrm{TS/T}). \tag{3}$$

Here r_b is the PSHELL bending-ratio field ($12I/T^3$, default 1.0), and k_s is the transverse-shear factor (PSHELL field TS/T, default 5/6). The drill stiffness is added separately through $\alpha_{\text{drill}} = (k_6/10^5)\, G\, h$ with k_6 from PARAM,K6ROT; see §12.

Membrane-only and bending-only PSHELLs. PSHELL permits MID2 (bending) to be blank, which Reference interprets as "no bending stiffness". The parser detects that case at src/parsing/extract_properties.jl:9-14, stores BEND_RATIO=0.0, and the kernel collapses $\mathbf{C}_b = \mathbf{0}$. In that branch the assembly path also forces $k_s \cdot \mathbf{C}_s = 0$ and $\alpha_{\text{drill}} = 0$: a pure membrane shell. Conversely a blank MID1 (membrane) is treated as "no membrane stiffness" and the element is bending-only. These two limits are detected directly from the property dictionary and require no special user action.

4.2 Anisotropic MAT2

MAT2 carries a full 3×3 symmetric in-plane stiffness ($G_{11}, G_{12}, G_{13}, G_{22}, G_{23}, G_{33}$). The parser builds $\mathbf{C}_m = h\,\mathbf{G}$ and $\mathbf{C}_b = (r_b h^3/12)\,\mathbf{G}$. Transverse shear is taken from the MID3 link if present, otherwise defaulted from (E, ν) as in the isotropic case. MAT2 is documented as *limited*: the parser also derives an equivalent (E, G, ν) tuple (src/ModelBuilder.jl:1000-1010) so downstream paths that expect scalar material parameters keep working.

4.3 Orthotropic MAT8

MAT8 is the standard Reference orthotropic 2-D material. The parser reads $(E_1, E_2, \nu_{12}, G_{12}, G_{1Z}, G_{2Z})$. The plane-stress reduced stiffness Q_{ij} is computed and rotated by the per-element angle θ (CQUAD4/CTRIA3 THETA/MCID field) so the per-element $\mathbf{C}_m$ and $\mathbf{C}_b$ reach the kernel already in the element frame. The relevant block is src/solver/assembly.jl:2382-2402; the standard rotation formulas for $Q_{11}, Q_{12}, Q_{22}, Q_{66}, Q_{16}, Q_{26}$ are implemented inline.

Rigid transverse-shear limit. A subtle but important detail of MAT8 parity is that Reference treats blank G_{1Z}/G_{2Z} as *rigid* transverse shear (i.e. Kirchhoff limit) following the convention of the Reference theoretical manual [14], whereas a literal Whitney–Pagano κ correction would *soften* it. JFEM tracks this convention at the parser level via the flag TRANSVERSE_SHEAR_RIGID_LIMIT on the shell property and applies the override at src/ModelBuilder.jl:615-648; the default behaviour for a qualifying PCOMP that uses MAT8 plies with blank G_{1Z}/G_{2Z} is $\mathbf{C}_s = 2.5 \cdot \mathbf{C}_s^{\text{Whitney}}$. The two escape hatches are documented in §12: setting JFEM_PCOMP_RIGID_TS_DISABLE=true restores the bare Whitney–Pagano stiffness, and JFEM_PCOMP_RIGID_TS_CS_SCALE replaces the 2.5 factor with a user-chosen value. The 2.5 default is deliberately conservative: it was tuned on the airframe launch-load validation set to maximise the mean MAC against Reference without over-stiffening pure flat-plate cantilevers, and it replaces an earlier 100× override that pushed mode-1 eigenvalues high by $\sim 15\%$ on curved aerospace shells.

4.4 Composite PCOMP via classical laminate theory

PCOMP is the most elaborate shell-property path. The model builder expands the ply stack (with the LAM flag controlling symmetric/antisymmetric layouts), rotates each ply by its own θ_{ply} angle, and assembles the standard classical laminate-theory (CLT) matrices [13]

$$\mathbf{A}_{ij} = \sum_k \bar{Q}_{ij}^{(k)}\,(z_k - z_{k-1}), \qquad \mathbf{B}_{ij} = \tfrac{1}{2}\sum_k \bar{Q}_{ij}^{(k)}\,(z_k^2 - z_{k-1}^2), \tag{4}$$

$$\mathbf{D}_{ij} = \tfrac{1}{3}\sum_k \bar{Q}_{ij}^{(k)}\,(z_k^3 - z_{k-1}^3), \qquad \mathbf{A}_s = \sum_k \mathbf{G}_{\text{sh}}^{(k)}\,(z_k - z_{k-1}). \tag{5}$$

The resulting matrices are stored on the shell property as $\mathbf{C}_m = \mathbf{A}$, $\mathbf{C}_b = \mathbf{D}$, $\mathbf{B}_{mb} = \mathbf{B}$ (or nothing when $\|\mathbf{B}\|$ is below a tolerance, i.e. a symmetric laminate), and $\mathbf{C}_s = \text{TS_T} \cdot$

$\mathbf{A}_s$ with the rigid-TS override described above when applicable. The relevant code is around `src/ModelBuilder.jl:560-677`. The unmodified $\mathbf{C}_s$ is kept as `Cs_raw` for diagnostic use.

Per-ply data. Each ply carries $(t, \theta, \mathbf{Q}, \rho)$ plus its position z_{k-1}, z_k, and the stack is retained on the property as `PLY_DATA`. This list is consumed by the MITC4-3D ply-by-ply path (§6.4) and by ply-level stress recovery (§10). The recovery formula at top/bottom of each ply uses the rotated $\mathbf{Q}^{(k)}$ and the recovered laminate mid-surface strains.

Equivalent scalar material. A synthetic equivalent `MAT1` is built per `PCOMP` as $E_{\text{eq}} = A_{11}(1 - \nu_{\text{eq}}^2)/h$, $\nu_{\text{eq}} = \min(A_{12}/A_{11}, 0.49)$, $G_{\text{eq}} = \frac{1}{2}(A_{s,11} + A_{s,22})/h$. This is used by downstream procedures that expect scalar material constants (drill stiffness, mass-density propagation, gravity) and is stored under a synthetic `MID = 900000+PID`.

Effective density. The effective density for gravity is $\rho_{\text{eff}} = (\sum_k \rho^{(k)}\, t^{(k)})/h$, computed in the same block.

Membrane–bending coupling $\mathbf{B}_{mb}$. When $\|\mathbf{B}\| > 10^{-10}\,\|\mathbf{A}\|$ the coupling matrix is attached to the property as $\mathbf{B}_{mb} = \mathbf{B}$. Several specialised shell branches (DKQ, DKMQ, Adini, flat-isotropic exact-membrane) are only activated when $\mathbf{B}_{mb}$ is absent (symmetric laminate), so the coupling test is also a routing test in the assembly dispatcher; see §5.

4.5 Thermal coupling: `MATT1` and `TABLEM1`

`MATT1` expresses temperature dependence of `MAT1` through scalar $E(T)$, $G(T)$, $\nu(T)$, $\rho(T)$, $\alpha(T)$ tables (`TABLEM1`). For shells, JFEM threads `TEMP(LOAD)` subcase data through the same isotropic stiffness and mass kernels, so that the same line, shell and solid operators are reassembled under the active subcase temperature field before the linear, modal or buckling solve. This is consistent with the short description in the previous manuscript and is implemented in `src/solver/assembly.jl` alongside the line and solid branches.

4.6 `PSHELL` field routing

The full `PSHELL` record is normalised in `src/parsing/extract_properties.jl`. The relevant fields and how they reach the shell kernel are summarised in Table 1.

Table 1: `PSHELL` fields consumed by the shell kernel.

Field	Default	Role
`MID1`	—	Membrane material id; selects $\mathbf{C}_m$ (`MAT1`, `MAT2`, `MAT8`).
`T`	—	Membrane thickness h. Drives all thickness powers in $\mathbf{C}_m, \mathbf{C}_b, \mathbf{C}_s$.
`MID2`	blank	Bending material id. Blank ⇒ membrane only; bending stiffness forced to zero.
`12I/T^3`	1.0	Bending ratio r_b. Scales $\mathbf{C}_b$; values < 1 model thin-plate softening, > 1 model effective stiffening.
`MID3`	blank	Transverse-shear material. Blank ⇒ default to $\mathbf{C}_s$ from E, ν or the MAT8 ply-aware path.
`TS/T`	5/6	Transverse-shear factor k_s.
`NSM`	0.0	Non-structural mass per unit area. Added to the consistent mass matrix.

4.7 `PSHEAR` and `CSHEAR`

`PSHEAR` is converted in the model builder into a synthetic `PCOMP_CLT` record with only the in-plane shear component G_{xy} non-zero (so $\mathbf{C}_m$ has a single non-zero entry at the (γ_{xy}) row, $\mathbf{C}_b = \mathbf{0}$,

$\mathbf{C}_s = \mathbf{0}$). The CSHEAR element then goes through the same quad pipeline. This keeps the assembly path uniform without needing a dedicated shear-panel kernel.

5 The CQUAD4 dispatcher

src/solver/assembly.jl contains a single per-element dispatcher that picks one of the kernels described in §6. The decision uses six categories of input:

1. **Property type**: TYPE field of the shell property (PSHELL or PCOMP_CLT).
2. **Material classification**: scalar isotropic, MAT8 orthotropic, MAT2 anisotropic, or PCOMP laminate. PCOMP carries an IS_ISOTROPIC flag for stacks whose plies all share E, G, ν; those laminates can fall through to the isotropic branches.
3. **Geometry**: planarity (warp deviation), aspect ratio, edge skew, axis-alignment (for the rectangular Adini branch), and the estimated principal curvature recovered from SNORM or from the geometric corner-normal field.
4. **Coupling**: whether $\mathbf{B}_{mb}$ is present (asymmetric laminate), whether PARAM,SNORM normals are uniform across the element, and whether the element is interior or isolated in its property cluster (used by the MacNeal bending-scale heuristics).
5. **Solution sequence**: a different branch may be selected for SOL 105 eigenvalue assembly than for SOL 101 linear-static assembly, because shell-thickness-locking sensitivity differs between the two operators. The dispatcher reads JFEM_Q4_KERNEL_STATIC and JFEM_Q4_KERNEL_EIG (and a unified JFEM_Q4_KERNEL fallback).
6. **Internal gates**: an extensive list of JFEM__* environment variables and PARAM_* model keys (§12). These can force, suppress or scale any branch.

The decision tree is applied per element, in the order listed in Table 2. The first row whose condition matches wins. The "Default" row at the bottom is the unconditional fallback.

In addition to the kernel choice, four *augmentations* are layered on top of the per-element $\mathbf{K}_e$ before assembly into the global matrix: (i) MacNeal residual-bending and twist corrections, (ii) Marguerre rotation–membrane coupling, (iii) static-component scaling for tagged elements (PID/EID filters, neighbour-prefix cluster guard, JFEM_Q4_STATIC_COMPONENT_*_SCALE family), and (iv) the drill stiffness α_{drill} from PARAM,K6ROT (with several override paths described in §12).

6 CQUAD4 kernels

Each subsection below describes one kernel: kinematics, B-matrices, integration rule, materials it accepts, and the gates relevant to it. The kernels live in src/FEMKernels.jl and are entered from src/solver/assembly.jl through a small set of stiffness_quad4_* factory functions.

6.1 The bilinear MITC4 default kernel

The default kernel is the four-node, bilinear, Reissner–Mindlin CQUAD4 of Dvorkin and Bathe (MITC4) [1,3] with assumed natural-coordinate transverse-shear strains, plus a Hughes–Brezzi drilling penalty for θ_z [6]. The two entry points, stiffness_quad4 and stiffness_quad4_generic, build the isotropic constitutive blocks and dispatch to stiffness_quad4_default_generic (verified at src/FEMKernels.jl:1326-1506) or to the full-feature stiffness_quad4_matrices (FEMKernels.jl:2948).

Table 2: `CQUAD4` dispatch order. The first matching condition wins. Conditions are intersected with their gates and per-element geometry tests.

#	Branch	Condition (summary)
1	MITC4-3D (§6.4)	`JFEM_Q4_KERNEL_*` requests `mitc4_3d` (unconditional) or `mitc4_3d_aspect` plus the aspect/warp/skew/curvature gates. With `ply_integration` on and a `PCOMP`, integration is per ply.
2	Curved-iso shear-centre-only blend	`SOL 105` eigen, isotropic, element flagged for shear-centre-only treatment; a centre-only $\mathbf{B}_s$ is blended with the full 2×2 MITC4 shear by `JFEM_CURVED_ISO_EIG_FULLSHEAR_BLEND`.
3	Flat DKMQ plate (§6.6)	Element flat, orthotropic/PCOMP, $\mathbf{B}_{mb}$ absent, $\mathbf{C}_b \neq 0$, DKMQ gate on.
4	Rectangular Adini plate (§6.7)	Element flat *and* axis-aligned rectangle, plate branch active, Adini gate on.
5	Flat DKQ plate (§6.5)	Element flat, orthotropic/PCOMP, $\mathbf{B}_{mb}$ absent, $\mathbf{C}_b \neq 0$, DKQ gate on or auto-detected.
6	PCOMP shear-centre-only / curved-PCOMP blend	`SOL 105` eigen, PCOMP, curved element; centre-only and full operators blended by `JFEM_CURVED_PCOMP_EIG_FULLSHEAR_BLEND`.
7	Flat-iso exact-membrane / flat-curved iso / cylindrical iso	Various curvature regimes; replaces the bilinear membrane operator by a covariant one tied to the geomnormal frame.
8	PCOMP exact-membrane	Flat orthotropic PCOMP with $\mathbf{B}_{mb} = \mathbf{0}$ and uniform `SNORM`; covariant membrane plus DKMQ/DKQ bending.
9	Default (§6.1)	Bilinear MITC4 with ϕ_2 anti-locking, optional MacNeal RBF, optional Wilson incompatible modes, optional MITC4+ assumed strain.

Membrane B. Standard bilinear shape functions $N_a(\xi,\eta) = \frac{1}{4}(1 \pm \xi)(1 \pm \eta)$ are used. At each 2×2 Gauss point the membrane B-matrix is built directly from $\partial N_a/\partial x$, $\partial N_a/\partial y$. Two enhancements can be turned on, controlled by the `membrane_incomp` and `membrane_assumed_mode` keyword arguments:

- Wilson incompatible modes: $\phi_1 = (1-\xi^2)$ and $\phi_2 = (1-\eta^2)$ as four bubble functions, statically condensed via $\alpha = -\mathbf{K}_{bb}^{-1}\mathbf{K}_{ab}^{\top}u$. Used as the default for static assembly and gated off for the legacy "Reference-bilinear" comparison (`JFEM_SOL105_STATIC_MEMBRANE_INCOMP=false`).
- MITC4+ assumed natural strain [4]: a Ko/Lee/Bathe-type assumed natural-coordinate membrane field, available in two variants (`:mitc4plus_linear_J`, `:mitc4plus_quadratic_J`) and ungated by default for flat elements; the curved-shell variant is held behind the global `MITC4PLUS_ALLOW_CURVED` switch (`FEMKernels.jl:17-19`).

Bending B. Standard C^0 rotations θ_x, θ_y are sampled at the same 2×2 Gauss points. A MacNeal twist correction is optionally applied at the Gauss-point level, replacing the twist row of $\mathbf{B}_b$ by $2\,\mathbf{B}_b^{\mathrm{gp}} - \mathbf{B}_b^{\mathrm{centre}}$ to cancel the $\xi\eta$ bending mode that otherwise locks.

Shear B with MITC4 tying. The kernel evaluates the covariant transverse-shear strains at the four MITC4 tying points $(\xi,\eta) \in \{(0,\pm1),(\pm1,0)\}$ (corresponding to the midpoints of the four element edges), constructs the tying-point $\mathbf{B}_s$ rows, then interpolates them with the bilinear shape functions back to each 2×2 Gauss point. The covariant strains are converted to physical strains by $\mathbf{C}_s^{\mathrm{phys}} = (J^{-1})^{\top}\,\mathbf{C}_s\,J^{-1}$ where J is the centre Jacobian, and multiplied by an anti-locking factor

$$\phi_2(\xi,\eta) \;=\; \min\Big(1,\; \alpha\,\frac{h^2}{4\,|\det J(\xi,\eta)|}\Big), \tag{6}$$

where α defaults to 10 (the global `PHI2_ALPHA Ref`, `FEMKernels.jl:10`). The factor is tuned to reproduce the Bathe–Dvorkin behaviour on thin shells while cancelling shear locking on coarse meshes; it is a covariant analog of the stress-projection remedies of Belytschko et al. [16]. Setting $\alpha = 0$ disables the correction.

Drilling term. The rotational stiffness around $\mathbf{n}$ is added as a penalty

$$\alpha_{\text{drill}} \;=\; (k_6/10^5)\, G\, h, \tag{7}$$

with k_6 from `PARAM,K6ROT` (default 100, Reference default). Several gates can override k_6 per element (see §12); the most common one is the rule "$k_6 = 0$ when the bending ratio is zero (membrane-only)", which is applied unconditionally so that AUTOSPC sees a pure membrane mechanism.

Quadrature. A 2×2 Gauss rule on (ξ, η). The in-plane stiffness contributions accumulate as $\mathbf{K}_e \mathrel{+}= w_g\, |\det J|\, B^\top D B$. The shear contribution uses the same Gauss rule with the tied shear rows. Incompatible modes are condensed after the full Gauss loop.

6.2 The MacNeal residual-bending kernel

For thin elements with low warp the kernel switches into a different shear treatment: the MacNeal 1978 residual-bending-flexibility correction [5]. At four shear sample points the kernel builds the covariant shear rows in the same way as MITC4, but the elemental shear stiffness is then replaced by the MacNeal formula

$$\tilde{\mathbf{B}}_s \;=\; \mathbf{B}_s^{\text{MITC4}} \;+\; \frac{L^2}{12\, EI}\, \mathbf{1}, \tag{8}$$

implemented through an explicit 4×4 residual-flexibility matrix Z_b and inverted in closed form. The default zb-scale is 0.65 (set in 2026-05-14; the legacy value 1.0 is recoverable via `JFEM_Q4_MACNEAL_RBF_ZB_SCALE_LEGACY=true`). The MacNeal twist correction described above is applied jointly. Eligibility for the MacNeal kernel is gated by a warp tolerance (`JFEM_Q4_MACNEAL_WARP_TOL`, default 10^{-4} relative) and an aspect-ratio cap (`JFEM_Q4_MACNEAL_ASPECT_MAX`, effectively unbounded by default).

The MacNeal path also exposes a family of bending-scale knobs: `JFEM_Q4_MACNEAL_BENDING_SCALE` for a uniform multiplier, `JFEM_Q4_MACNEAL_BENDING_ISOLATED_SCALE` for isolated elements (valence 1), `JFEM_Q4_MACNEAL_CURVED_BENDING_SCALE` for the curved-shell regime, and two aspect-ratio interpolation series (`_ASPECT_SCALE` and `_ASPECT2_SCALE`) that allow fine calibration of the MacNeal bending response on highly elongated cells. These are recorded in detail in §12.

6.3 MITC4+ assumed natural-strain membrane

The default kernel can be augmented with the Ko–Lee–Bathe 2017 MITC4+ assumed-strain membrane field [4]. The `membrane_assumed_mode` keyword selects the variant: `:mitc4plus_linear_J` or `:mitc4plus_quadratic_J`. The flat-only variant is always available; the curved variant is gated by the module-level `MITC4PLUS_ALLOW_CURVED` switch (`FEMKernels.jl:12-19`), which reads `JFEM_SOL105_EIG_MITC4PLUS_ALLOW_CURVED`. The motivation is to mitigate membrane locking on distorted four-node shells (the tapered lifting-surface family in the airframe validation set).

6.4 The MITC4-3D degenerate-shell kernel

For curved or warped elements that the dispatcher routes through the `mitc4_3d` or `mitc4_3d_aspect` branches, the kernel is the 3-D Dvorkin–Bathe degenerate shell [2]. The element is

parameterised in (ξ, η, ζ) with $\zeta \in [-1, 1]$ the through-thickness coordinate, the position field is

$$\mathbf{x}(\xi, \eta, \zeta) \;=\; \sum_a N_a(\xi, \eta) \left(\mathbf{x}_a + \tfrac{h}{2}\, \zeta\, \hat{\mathbf{V}}_a \right), \tag{9}$$

where $\hat{\mathbf{V}}_a$ is the nodal director, taken from `SNORM` or from the averaged geometric normal at each node. The covariant metric is built directly from the 3-D Jacobian, the contravariant shear strains are tied at the four MITC4 points, and the kernel returns a 24×24 element stiffness in the global frame. The relevant entry points are `stiffness_quad4_mitc4_3d_resultant_matrices` (`FEMKernels.jl:658-796`) for the laminate-resultant path and `stiffness_quad4_mitc4_3d_ply_matrices` (`FEMKernels.jl:798`) for the ply-by-ply integration path used with `PCOMP`.

Internal switches for this kernel are exposed as `JFEM_Q4_MITC4_3D_MEMBRANE_SCALE`, `JFEM_Q4_MITC4_3D_BENDING_SCALE`, `JFEM_Q4_MITC4_3D_SHEAR_SCALE` and `JFEM_Q4_MITC4_3D_DRILL_SCALE` (each defaulting to 1.0), plus a shear-mode selector `JFEM_Q4_MITC4_3D_SHEAR_MODE` that can choose `covariant` (default), `physical` or `macneal/rbf`. A bending perturbation `JFEM_Q4_MITC4_3D_BENDING_ZETA_DELTA` (default 10^{-4}) controls the numerical differentiation of $\mathbf{C}_b$ from $\mathbf{C}_m$ through the shell thickness in the resultant path.

The MITC4-3D path is the recommended next development direction for closing the largest remaining parity gaps against Reference on curved-shell buckling (see Section 14). It is presently *not* the default kernel; switching to it is an opt-in choice through `JFEM_Q4_KERNEL_*`.

6.5 The discrete-Kirchhoff DKQ plate kernel

When the element is flat, $\mathbf{B}_{mb}$ is absent, and $\mathbf{C}_b \neq 0$, the dispatcher can pick the DKQ plate kernel, in the discrete-Kirchhoff family introduced for triangles in [7] and extended to quadrilaterals by Batoz and Tahar (1982). Transverse shear is enforced to vanish at the four edge midpoints, the rotations are interpolated by quadratic edge functions, and the plate bending is integrated by 2×2 Gauss. The kernel only carries the plate DOFs (w, θ_x, θ_y); the membrane block comes from the bilinear path unchanged. The implementation lives at `FEMKernels.jl:3854-4039,5301-5348` and is exposed through `stiffness_quad4_plate_dkq_matrices`.

DKQ has no transverse-shear deformability (it is a thin-plate kernel) and is therefore restricted to plates whose shear is negligible. The dispatcher's flat-plate gate requires both *plate-branch enabled* and *flat-orthotropic/PCOMP with* $\mathbf{B}_{mb} = \mathbf{0}$; otherwise it falls through to the DKMQ or default kernel.

6.6 The discrete-Kirchhoff–Mindlin DKMQ plate kernel

DKMQ [10] extends DKQ to include transverse shear in a way that preserves the discrete-Kirchhoff constraints, with the membrane improvements of Stembera and Füssl [11] available as a future development direction. Per-edge hierarchical rotation locking is introduced through

$$\kappa_k \;=\; \frac{12}{L_k^2} \Big((\mathbf{H}_s^{-1})_{2,1}\, (\mathbf{H}_b)_{3,2} + (\mathbf{H}_s^{-1})_{2,2}\, (\mathbf{H}_b)_{2,2} \Big), \tag{10}$$

with the orthotropic/PCOMP-aware $\mathbf{H}_b, \mathbf{H}_s$ blocks constructed at the side. The implementation is at `FEMKernels.jl:4057-4126,4170-4261,5349-5378`. `stiffness_quad4_plate_dkmq_matrices` is the entry point and `add_quad4_plate_dkmq_exact_shear!` adds the exact side-shear operator to an existing element stiffness.

The DKMQ kernel is, in practice, the preferred plate kernel for flat orthotropic PCOMP. It cleanly handles bending–twist coupling on unbalanced laminates and is the default when the flat-DKMQ gate (`JFEM_SOL105_EIG_FLAT_PCOMP_DKMQ`, default `true`) is on.

6.7 The Adini rectangular plate kernel

For axis-aligned rectangular elements the dispatcher can choose the Adini 1961 plate (full-cubic Hermite deflection $w(x, y)$, with corner DOFs $(w, \partial_x w, \partial_y w)$). The kernel is *not* a Reissner–Mindlin kernel and carries no transverse shear; it is integrated analytically through a 4×4 Gauss–Legendre rule on the bending block. The membrane block comes from the standard MITC4 path. The implementation is at `FEMKernels.jl:4894-5070` with `stiffness_quad4_plate_adini_matrices` as the entry point.

Adini is restricted to flat axis-aligned rectangles by the geometry test (`FEMKernels.jl:4957`). Out-of-bounds elements fall through to DKQ. The branch is principally useful for clean rectangular plate benchmarks where the analytical cubic field minimises discretisation error.

6.8 The Tessler–Hughes MIN4 anisoparametric kernel

The MIN4 kernel of Tessler and Hughes [20] is an anisoparametric plate kernel with biquadratic out-of-plane deflection and bilinear rotations, with the transverse-shear shape functions constrained to drop the spurious zero-energy mode. It is activated by `JFEM_Q4_KERNEL=min4` (or `tessler_hughes`, `tessler-hughes`) and is implemented at `FEMKernels.jl:4263-4449` via `stiffness_quad4_min4_bending_shear`. The membrane block reuses the standard bilinear path with $\mathbf{C}_b = \mathbf{C}_s = 0$.

The MIN4 kernel exposes a single calibration constant `cbmin4`, defaulted to 3.6 to match the MYSTRAN MIN4 reference, and tunable through `JFEM_MIN4_CBMIN4`. The kernel is currently a research switch: useful for cross-checks against MYSTRAN but not the default for any `SOL` branch.

6.9 Hybrid-stress (Pian–Sumihara) variants

`FEMKernels.jl:2058-2327` implements three hybrid-stress variants: a membrane-only block (`stiffness_quad4_membrane_hybrid_stress_matrices`), a bending-only block (`stiffness_quad4_bending_hybrid_stress_matrices`), and a combined membrane–bending block (`stiffness_quad4_membrane_bending_hybrid_stress_matrices`).

Each variant assumes a five-mode stress field in J_0-transformed coordinates and assembles

$$\mathbf{K}_e \;=\; \mathbf{G}^{\top}\mathbf{H}^{-1}\mathbf{G}, \qquad \mathbf{H} \;=\; \int P^{\top}\mathbf{S}^{-1}P \,\mathrm{d}A, \qquad \mathbf{G} \;=\; \int P^{\top}B \,\mathrm{d}A, \tag{11}$$

through 2×2 Gauss. The hybrid-stress kernels are currently in research mode and are exercised through targeted regressions; they are not the default for any solution sequence.

6.10 Hu–Washizu mixed kernel

`stiffness_quad4_huwashizu_matrices` (`FEMKernels.jl:2873-2947`) implements the three-field Hu–Washizu variational formulation (displacement, strain, stress); see [12] for a recent four-node Cosserat-shell variant of the same family with the same five hybrid-stress modes plus a parallel five-mode strain field. It is gated by `JFEM_Q4_KERNEL=huwashizu` (also `hu-washizu`, `hw`). The full 24×24 element stiffness is assembled without condensation. The Hu–Washizu kernel is a research kernel that bypasses the incompatible-mode, MITC4+ and curvature heuristics; it is useful as a clean independent baseline against which the other paths can be benchmarked.

6.11 Membrane stress-recovery operators

Several stand-alone operators support membrane recovery for $\mathbf{K}_G$and stress output:

- `quad4_membrane_force_field` (`FEMKernels.jl:5994`) — 2×2 Gauss-point membrane resultants (N_{xx}, N_{yy}, N_{xy}) with optional incompatible-mode condensation.

- `quad4_membrane_force_field_triangle_recovery` (`FEMKernels.jl:6301`) — corner-resultant recovery via decomposition of the quad into four constant-strain triangles, with averaging modes (`:tri_aspect`, `:tri_center_adj`, `:tri_incident_interp`, `:tri_diagavg`).
- `quad4_membrane_force_field_covariant` (`FEMKernels.jl:6690`) — covariant resultants for curved shells in 3-D coordinates.
- `quad4_membrane_cst_resultant` (`FEMKernels.jl:6194`) — exact constant-strain-triangle resultant from a chosen node triple.

These operators feed the *stress-field classifier* of §9, which picks between Gauss-point, average and covariant inputs to the geometric-stiffness operator.

7 The `CTRIA3` triangle

JFEM does not implement a dedicated triangular shell kernel. The alternatives in the literature (the DKT family of Batoz–Bathe–Ho [7], the simple 6-DOF triangular shell of Bathe and Ho [8], the MITC3+ element of Lee–Jeon–Lee–Bathe [9]) are all viable but none is presently exercised; instead the `CTRIA3` is built as a macro-quad superposition: each triangle is split into three overlapping quadrilaterals that share a virtual centroid node, the three quads are assembled with the standard `CQUAD4` kernel of §6.1, only the plate DOFs are kept (rows/columns $3, 4, 5, 9, 10, 11, 15, 16, 17, 21, 22, 23$ of the 24×24 element matrix, see `QUAD4_PLATE_DOF_IDX` at `FEMKernels.jl:6783`), and the seventh (centroidal) node is statically condensed.

Geometry of the macro-quad split. The triangle's three corner nodes are augmented by three edge midpoints and one centroid. The three macro-quads share that centroid: `TRIA3_MACRO_QUADS` $= ((1, 4, 7, 6), (2, 5, 7, 4), (3, 6, 7, 5))$ (`FEMKernels.jl:6782`), with node indices over the seven virtual points (`FEMKernels.jl:6785-6802`).

Condensation. After assembling $\mathbf{K} \in \mathbb{R}^{21\times 21}$ from the three macro-quads, the external 9 DOFs are kept and the internal 12 DOFs (centroidal node plus the three midpoints) are condensed via $\mathbf{K}_{\text{cond}} = \mathbf{K}_{ee} + \mathbf{K}_{ei}\,\mathbf{A}_{\text{int}}$ where $\mathbf{A}_{\text{int}} = -\mathbf{K}_{ii}^{-1}\,\mathbf{K}_{ie}$. The condensation map $\mathbf{A}_{\text{int}}$ is retained so that pressure loads, stress recovery and geometric stiffness can later push the internal DOFs back through the same partition. See `tria3_plate_macro_data` at `FEMKernels.jl:6810-6862` for the implementation.

Materials. The triangle accepts the same $(\mathbf{C}_m, \mathbf{C}_b, \mathbf{C}_s, \mathbf{B}_{mb})$ data as the quad, so all four material categories (`MAT1`, `MAT2`, `MAT8`, `PCOMP`) flow through unchanged.

Drill and bending-only modes. The triangle obeys the same membrane-only/bending-only conventions as the quad: a `PSHELL` with blank `MID2` forces $\mathbf{C}_b = \mathbf{0}$ and zeroes the drill stiffness. The implementation early-exits at `FEMKernels.jl:6817` when the bending ratio is below 10^{-12}.

Pressure load. Pressure loads on triangles (`tria3_plate_macro_pressure_load` at `FEMKernels.jl:6864`) build a 12-vector on the macro-quad, condense through the same $\mathbf{A}_{\text{int}}$ map, and return the 9-vector consistent with the condensed stiffness.

8 Higher-order shells: `CTRIA6` and `CQUAD8`

JFEM does not yet provide quadratic shell kernels. `CTRIA6` and `CQUAD8` cards are accepted by the parser and expanded during assembly into lower-order sub-elements:

- `CTRIA6`: each input is replaced by four `CTRIA3` sub-elements, one per corner region plus a central triangle: $(1,4,6),(4,2,5),(5,3,6),(4,5,6)$, where $4,5,6$ are the edge midpoints. Element ID, `PID`, `THETA`, `MCID` and any pressure loading are inherited.
- `CQUAD8`: each input is replaced by six `CTRIA3` sub-elements: $(1,5,8),(5,2,6),(6,3,7),(7,4,8),(5,6,8),(6,7,8)$.

The two decompositions are applied unconditionally during `assemble_stiffness` (`src/solver/assembly.jl` around the "Expanded $\langle n \rangle$ CQUAD8 + $\langle n \rangle$ CTRIA6 into sub-elements" log line). The same expansion is reused for mass and geometric stiffness so the model stays internally consistent. This preserves input compatibility at modest implementation cost; the trade-off is that the quadratic basis is not exploited and the effective element order is one.

9 Shell geometric stiffness for `SOL 105`

The shell geometric stiffness $\mathbf{K}_G$ is built from the membrane prestress $\boldsymbol{\sigma}_m$ recovered from the static preload solve. It uses the same shape functions as the linear stiffness but couples through the $\boldsymbol{\sigma}_m\,(\nabla N \otimes \nabla N)$ kernel. Optional rotational contributions (the "rotation-gradient" branch) and principal transverse-shear coupling can be activated by environment variables.

9.1 Stress-field recovery modes

Each `CQUAD4` can supply $\boldsymbol{\sigma}_m$ to $\mathbf{K}_G$ in one of several forms, selected by `JFEM_KG_QUAD4_STRESS_FIELD_MODE` and `JFEM_KG_QUAD4_MEMBRANE_RECOVERY_MODE`:

- `:planar` (default, corner-interpolated): the legacy 2-D plane-stress path. Stresses are recovered at corners and bilinearly interpolated to the 2×2 Gauss points.
- `:gauss`: stresses are recovered at the Gauss points directly. This is the most expensive but the most accurate option for general loads.
- `:average`: element-averaged stress. Cheap and stable but coarse; used by the auto-load classifier below for compression-dominated subcases.
- `:auto`: the dispatcher picks between Gauss and average based on whether the element passes the `kg_quad4_stress_field_mode` classifier.
- `:covariant`: covariant recovery for curved shells. The default when `JFEM_SOL105_EIG_CURVED_JACOBIAN=true`.
- Several triangle-recovery modes (`:tri_*`) for sub-cell averaging.

9.2 The auto-load classifier

The auto-average mode reads the static load distribution and decides whether the prestress is "axial-compression dominated" enough to justify the cheaper averaged $\boldsymbol{\sigma}_m$. The decision is the conjunction of five tests: (i) per-element geometry test, (ii) compression-only test on the averaged principal stresses, (iii) a biaxial-compression test, (iv) a shear-resultant ratio below a small threshold (default 2%), and (v) a global load-direction test (subcase `LOAD` dominated by a single axis with the matching sign). The classifier is enabled by default; the family of `JFEM_KG_QUAD4_AUTO_AVG_LOAD_*` variables documented in §12 controls its sensitivity.

9.3 Principal transverse-shear coupling

The shell $\mathbf{K}_G$ can include a transverse-shear contribution derived from the principal-direction stress decomposition. This is gated by `JFEM_KG_SHELL_TRANS_MODE=principal`; the default is `:none`. Three scale factors apply per principal-direction contribution: `JFEM_KG_SHELL_PRINCIPAL_SHEAR_YY_FACTOR`, `..._XY_FACTOR` and `..._Z_FACTOR`. The branch is restricted to flat elements by default (`JFEM_KG_SHELL_PRINCIPAL_TRANSVERSE_FLAT_ONLY=true`).

9.4 Rotation-gradient $\mathbf{K}_G$

A small higher-order $\mathbf{K}_G$ contribution from the rotation gradient is available through `JFEM_KG_SHELL_ROT_GRAD_SCALE` (default 1). The autoscale variants `JFEM_KG_SHELL_ROT_GRAD_AUTO_ISO_SCALE` and `..._AUTO_PCOMP_SCALE` tie the contribution to a curvature threshold and a cylindricity ratio.

9.5 PCOMP N_{xy} shear scale

`JFEM_KG_SHELL_PCOMP_NXY_SCALE` multiplies the shear prestress that enters $\mathbf{K}_G$ on `PCOMP` shells. It supports an aspect-ratio-dependent interpolation (`_LOW`,`_HIGH`,`_ASPECT_LOW/_HIGH`) and a shear-dominance relaxation knob (`_SHEAR_DOM_RELAX`) so that pure-shear cases are not double-penalised.

9.6 Membrane-incompatible $\mathbf{K}_G$

If $\mathbf{C}_m$ in the linear stiffness was built with Wilson incompatible modes, $\mathbf{K}_G$ can mirror the same condensation by setting `JFEM_KG_MEMBRANE_INCOMP=true`. The default behaviour is to inherit whatever the linear stiffness used through `JFEM_KG_USE_COMPATIBLE_MEMBRANE_STRESS`, so that the buckling and static operators remain on the same footing.

9.7 Plate-only $\mathbf{K}_G$ variants

For DKMQ and DKQ-plate elements the geometric stiffness is built with a matching plate operator (`geometric_stiffness_quad4_plate_dkmq`, `geometric_stiffness_quad4_plate_dkq`, `FEMKernels.jl`:5537–5577) that uses the edge relation from the linear stiffness. The Adini plate has its own `geometric_stiffness_quad4_plate_adini` (`FEMKernels.jl:5072`) with a falls-through-to-bilinear path for non-axis-aligned elements.

10 Shell stress recovery

Stress recovery for shells is handled by `src/solver/stress_recovery.jl`, with the heavy work done by helper routines in `src/FEMKernels.jl`. The resultant-recovery conventions follow the linear stress-resultant theory of Simo, Fox and Rifai [19], in the sense that membrane and bending resultants are recovered directly from the integrated constitutive blocks rather than reconstructed from through-thickness stress integration. The recovered data are:

- Membrane resultants (N_{xx}, N_{yy}, N_{xy}) at element centre, Gauss points and corners, depending on subcase.
- Bending moments (M_{xx}, M_{yy}, M_{xy}) at the same locations.
- Transverse shear (Q_x, Q_y) at the element centre.
- Top and bottom surface stresses $(\sigma_{xx}, \sigma_{yy}, \sigma_{xy}, \sigma_{vM})$ for isotropic and orthotropic shells.

- Per-ply stresses at top/bottom of every ply for `PCOMP`, rotated back from the ply axes by the per-ply θ.

The implementation cores are `stress_strain_quad4` (`FEMKernels.jl:5656-5842`) for the quadrilateral and `stress_strain_tria3` (`FEMKernels.jl:7044`) for the triangle, with a `for_kg=true` switch that skips the incompatible-mode contribution when the recovered stress is going into $\mathbf{K}_G$ rather than into a printed result. The transverse-shear recovery at the centre uses `quad4_mitc4_center_shear_resultant` (`FEMKernels.jl:5580-5654`).

For the curved-PCOMP buckling subset the recovery is routed through the covariant operator (`quad4_membrane_force_field_covariant`) so the same metric is used in both the forward stiffness and the buckling RHS; this is one of the architectural improvements highlighted in the previous manuscript.

11 Consistent mass for shells

Shell mass is assembled with consistent (not lumped) shape-function quadrature:

- `consistent_mass_quad4` (`FEMKernels.jl:7857`) — 2×2 Gauss; translational entries proportional to ρh, rotational entries proportional to $\rho h^3/12$, drilling zero.
- `consistent_mass_tria3` (`FEMKernels.jl:7903`) — analytical evaluation on the constant-strain triangle: diagonal entry $\rho h A/6$, off-diagonal $\rho h A/12$.

Non-structural mass from `PSHELL/NSM` or `PCOMP/NSM` is added on the diagonal as a per-unit-area term. `PARAM,WTMASS` multiplies the final mass matrix as in Reference. `CONM2` contributions are added separately by the model-builder mass path.

12 Catalogue of internal gates

This section is the long catalogue of every gate the shell layer reads. The categories follow the kernel descriptions above. Each table lists the environment variable, its default, and a one-line description. Reading the tables in conjunction with Table 2 should be enough to predict the operator the solver will assemble for any combination of inputs.

12.1 Global behaviour

Table 3: Global shell defaults exposed as `Ref`s in `src/FEMKernels.jl`.

`Ref` or env var	Default	Effect
`PHI2_ALPHA`	10.0	Coefficient α in the ϕ_2 shear anti-locking factor of eq. (6). Setting 0 disables the correction.
`MITC4PLUS_ALLOW_CURVED`	`false`	When true, the MITC4+ assumed-strain membrane is allowed on curved shells; otherwise restricted to flat ones. Reads `JFEM_SOL105_EIG_MITC4PLUS_ALLOW_CURVED`.
`JFEM_Q4_KERNEL`	`macneal`	Default kernel: `macneal`, `macneal_all`, `mitc4_3d`, `mitc4_3d_aspect`, `min4`, `huwashizu`.
`JFEM_Q4_KERNEL_STATIC` / `_EIG` / `_KG`	inherits	Per-pass override, so the static $\mathbf{K}$, the eigenvalue $\mathbf{K}$ and the geometric stiffness $\mathbf{K}_G$ can use different kernels.

12.2 Curvature handling

Table 4: Curvature- and frame-related gates.

Variable	Default	Effect
JFEM_CURVED_ISO_EIG_FULLSHEAR_BLEND	1.0	Blend factor: 0 centre-only, 1 full 2×2 MITC4 shear on the curved-iso shear-centre-only branch.
JFEM_CURVED_PCOMP_EIG_FULLSHEAR_BLEND	1.0	Same for the curved-PCOMP shear-centre-only branch.
JFEM_Q4_CURVED_ISO_SQUARE_FULLSHEAR_BLEND_VALUE	—	Override value used when the element aspect ratio is below the matching threshold.
JFEM_Q4_CURVATURE_MEMBRANE_SCALE_EIG / _STATIC	1.0	Global scale on the curvature-membrane coupling term that enters $\mathbf{B}_m$ when SNORM is available.
JFEM_Q4_CURVATURE_FILTER_MODE_EIG / _STATIC	:principal	Curvature filter (:principal, :cylindrical, :gaussian).
JFEM_Q4_CURVATURE_CYL_RATIO_MAX_EIG / _STATIC	—	Maximum allowed cylindricity ratio to apply the curvature term.
JFEM_Q4_CURVATURE_RESOLUTION_MIN_EIG / _STATIC	—	Minimum element size for curvature activation.
JFEM_Q4_CURVATURE_RESOLUTION_FULL_EIG / _STATIC	—	Element size at which the curvature contribution reaches full weight.
JFEM_Q4_FLAT_CURVED_ISO_GEOMNORMAL_FRAME	—	Project the rotation frame onto the geometric/SNORM normal on slightly curved flat-isotropic elements.
JFEM_Q4_CURVED_ISO_GEOMNORMAL_FRAME	—	Same, for non-flat curved isotropic elements.
JFEM_Q4_FLAT_CURVED_ISO_NODAL_GEOMNORMAL_TRANSFORM	—	Apply the geomnormal projection at each node rather than just at the centre.
JFEM_Q4_STATIC_PCOMP_NODAL_GEOMNORMAL_TRANSFORM	—	Enable the nodal projection in static assembly for PCOMP.
JFEM_SOL105_EIG_CURVED_JACOBIAN	false	Switches the buckling assembly to the curved-Jacobian integration path (experimental).

Table 5: Flat-isotropic membrane and shear options for `SOL 105`.

Variable	Default	Effect
`JFEM_SOL105_EIG_FLAT_ISO_MEMBRANE_INCOMP`	`true`	Enable Wilson incompatible membrane modes on flat isotropic shells.
`JFEM_SOL105_MEMBRANE_INCOMP_CENTER_JACOBIAN`	`false`	Use the centre Jacobian for the bubble functions (instead of per-Gauss-point).
`JFEM_SOL105_EIG_FLAT_ISO_MEMBRANE_SHEAR_CENTER_ROW`	`false`	Use the covariant centre-row shear field on flat isotropic shells.
`JFEM_SOL105_EIG_FLAT_ISO_MEMBRANE_ASSUMED_MODE`	`:none`	Membrane assumed-natural-strain selector: `:none`, `:mitc4plus`, `:mitc4plus_all`.
`JFEM_SOL105_EIG_FLAT_ISO_DKMQ`	`true`	Allow the flat-DKMQ bending branch on flat isotropic shells.
`JFEM_SOL105_EIG_FLAT_ISO_FULLSHEAR_SELECTIVE_MODE`	`:none`	Selective reduced-integration shear mode on the flat-isotropic static branch.
`JFEM_SOL105_STATIC_MEMBRANE_INCOMP`	`false`	Enable Wilson modes on the SOL 105 static preload (separate from the SOL 105 eigen path).
`JFEM_SOL105_STATIC_MEMBRANE_INCOMP_AUTO_LOAD`	`false`	Auto-enable the bubble-augmented static $\mathbf{K}$ for negative-X-dominated compression subcases.

12.3 Flat-isotropic eigenvalue gates

12.4 Flat-PCOMP eigenvalue gates

12.5 Curved-isotropic membrane incompatibility

12.6 MacNeal RBF and MITC4-3D

12.7 Drilling DOF (K6ROT) gates

12.8 Static-component scaling

12.9 PCOMP rigid-TS limit

12.10 Marguerre rotation–membrane coupling

12.11 Geometric stiffness

12.12 Hotspot / cluster diagnostics

The same static-component family above doubles as the "hotspot" / localisation-filter diagnostic. The `JFEM_Q4_STATIC_COMPONENT_EID_FILTER` variable selectively applies $\mathbf{C}_m$, $\mathbf{C}_b$, $\mathbf{C}_s$, $\mathbf{B}_{mb}$ and drill scaling to one or more named element IDs, while `JFEM_Q4_STATIC_COMPONENT_NEIGHBOR_PID_PREFIXES` acts as a *cluster guard*, restricting the scaling to elements that are topologically adjacent to a named property cluster. These are used in practice to bracket which element or property family is responsible for a parity miss in `SOL 105`, without rebuilding the whole deck.

13 Adjoint sensitivities on shells

The previous paper already documents the full adjoint workflow. This section restricts itself to the shell-specific contractions. The state of the adjoint with respect to shells is:

- **shell_thickness** (h): `ForwardDiff`-backed on both `CTRIA3` and the default `CQUAD4` stiffness kernel through the standard `stiffness_quad4_default_generic`/`stiffness_tria3_`

Table 6: Flat-PCOMP membrane, shear and bending options for `SOL 105`.

Variable	Default	Effect
`JFEM_SOL105_EIG_FLAT_PCOMP_NO_PHI2`	auto	Disable the ϕ_2 membrane-rotation correction on flat PCOMP.
`JFEM_SOL105_EIG_FLAT_PCOMP_PLATE_BRANCH`	—	Enable the explicit DKQ plate branch.
`JFEM_SOL105_EIG_FLAT_PCOMP_PLATE_AUTO`	—	Auto-detect the plate branch on axis-aligned rectangles.
`JFEM_SOL105_EIG_FLAT_PCOMP_PLATE_AUTO_D16_RATIO_MAX`	—	Maximum D_{16}/D_{11} to trigger the auto plate branch.
`JFEM_SOL105_EIG_FLAT_PCOMP_PLATE_AUTO_SHEAR_RATIO_MAX`	—	Maximum shear-stiffness ratio to trigger the auto plate branch.
`JFEM_SOL105_EIG_FLAT_PCOMP_RECT_ADINI`	—	Use the Adini plate on rectangular flat PCOMP elements.
`JFEM_SOL105_EIG_FLAT_PCOMP_DKMQ`	`true`	Enable the DKMQ plate branch on flat orthotropic PCOMP.
`JFEM_SOL105_FLAT_PCOMP_DKMQ_STATIC`	—	Allow DKMQ in static assembly.
`JFEM_SOL105_EIG_FLAT_PCOMP_FULLSHEAR_SELECTIVE`	`false`	Selective shear treatment on flat PCOMP static.
`JFEM_SOL105_EIG_FLAT_PCOMP_SHEAR_SCALE`	1.0	Multiplier on $\mathbf{C}_s$ for SOL 105 eigen flat PCOMP.
`JFEM_SOL105_EIG_FLAT_PCOMP_AUTO_SHEAR_SCALE`	—	Compute the shear scale automatically from a laminate metric.
`JFEM_SOL105_EIG_FLAT_PCOMP_AUTO_SHEAR_SCALE_GAIN`	—	Gain on the auto shear scale.
`JFEM_SOL105_EIG_FLAT_PCOMP_AUTO_SHEAR_SCALE_MAX`	—	Upper limit for the auto shear scale.
`JFEM_SOL105_EIG_FLAT_PCOMP_EXACT_MEMBRANE`	—	Use the exact covariant membrane operator on flat PCOMP.
`JFEM_SOL105_EIG_FLAT_PCOMP_EXACT_SIDE_SHEAR`	—	Use the DKMQ exact side-shear operator.
`JFEM_SOL105_FLAT_PCOMP_EXACT_SIDE_ROTCORR`	—	Include the rotation correction in the exact side-shear path.
`JFEM_SOL105_EIG_FLAT_PCOMP_MEMBRANE_ASSUMED_MODE`	—	Membrane ANS selector for flat PCOMP.
`JFEM_SOL105_EIG_FLAT_PCOMP_TAPER_MEMBRANE_NONE`	—	Disable membrane modes on tapered flat PCOMP (high aspect ratio).

`matrices` paths.

- **material_NU**: `ForwardDiff`-backed on the same shell kernels; on the buckling side, $\partial \mathbf{K}_G/\partial\nu$ is evaluated exactly through $(\partial \mathbf{D}/\partial\nu)\mathbf{B}_m u$ on the standard and flat-specialised shell paths, with synchronised branch-aware local contractions on the verified covariant subset.
- **pcomp_ply_thickness** and **pcomp_ply_angle**: exact analytical differentiation of the stored CLT $(\mathbf{A}, \mathbf{B}, \mathbf{D}, \mathbf{A}_s)$ matrices, with a finite-difference fallback retained as a compatibility path.
- **node_coord**: full-model finite-difference reassembly after geometric perturbation. This is the most expensive shell adjoint route and is reserved for small mesh-shape studies.

For the buckling adjoint the SOL 105 shell paths covered analytically or through branch-aware local finite differences are: the standard MITC4 shell, the flat-isotropic specialised subset (with the exact shell-level $\partial\boldsymbol{\sigma}/\partial\nu = (\partial \mathbf{D}/\partial\nu)\mathbf{B}_m u$ contraction), the flat-curved cylinder-panel and barrel/tapered-cylinder families, the explicit covariant isotropic `CQUAD4` subset, the

Table 7: Curved-isotropic membrane-incompatibility gates.

Variable	Default	Effect
`JFEM_SOL105_EIG_CURVED_ISO_AUTO_MEMBRANE_INCOMP`	—	Auto-add Wilson bubble modes on curved-isotropic elements that pass the curvature gates.
`JFEM_SOL105_EIG_CURVED_ISO_AUTO_MEMBRANE_INCOMP_KAPPA_L_MIN`	—	Minimum curvature κL.
`JFEM_SOL105_EIG_CURVED_ISO_AUTO_MEMBRANE_INCOMP_CYL_RATIO_MAX`	—	Maximum cylindricity.
`JFEM_SOL105_EIG_CURVED_ISO_WARP_MEMBRANE_INCOMP`	—	Same, on warped curved-iso elements; tied to a warp threshold and a curvature cap.
`JFEM_SOL105_EIG_CURVED_ISO_ELONGATED_MEMBRANE_INCOMP`	—	Same, on elongated curved-iso elements; tied to an aspect-ratio floor.

compatible-membrane auto-covariant subset, and the flat orthotropic `PCOMP` family with branch-aware DKMQ/DKQ selection. The hardest exact-membrane curved-shell groups still fall back to a mode-tracked full-model finite difference for the complete eigenvalue derivative; the local cached and full-model paths are reported separately in the `.ADJOINT.JSON` diagnostics, exactly as described in the previous manuscript.

14 Verification and current limits relevant to shells

Broader validation uses independent reference comparisons, element probes, and archived numerical results. The public repository carries publishable shell checks under `validation/cases/`. From the shell-specific perspective, the open items are:

- **Curved `PCOMP` membrane/shear-to-rotation coupling**: the strong-curvature lifting-surface strain-driven family is the currently dominant gap (MAC= 0.0 on subcase 511002). The component-removal diagnostic (`JFEM_Q4_STATIC_COMPONENT_CS_SCALE`, etc.) shows the problem is not a scalar shear multiplier but a shape issue in the curved-shell rotation columns. The planned next step (documented in `MITC4_3D_SHELL_IMPLEMENTATION_PLAN.md`) is to promote the MITC4-3D degenerate-shell kernel of §6.4 from an opt-in alternative to the default, once the per-deck acceptance criteria pass.
- **Membrane recovery on curved cylindrical patches**: the Reference-u FORCE-resultant probe shows that the recovery equations themselves are sound; the residual is upstream in the static curved-PCOMP stiffness shape. The current covariant recovery operator closes most of the gap when the curved-Jacobian gate (`JFEM_SOL105_EIG_CURVED_JACOBIAN`) is on.
- **Higher-order shells**: `CTRIA6`/`CQUAD8` are served by subdivision, not by a dedicated kernel. A native quadratic shell is a candidate development item but is not scheduled.
- **SOL 106 on shells**: the experimental nonlinear branch is limited to flat coplanar `CTRIA3`/`CQUAD4` `PSHELL` decks. The von Kármán internal-force `f_vk(u)` and its consistent tangent $\partial \mathbf{f}_{\mathrm{vk}}/\partial u$ are presently restricted to this subset.
- **Drilling on triangles**: the triangle inherits the same α_{drill} as the quad through the macro-quad decomposition; an independent triangular drill formulation has not been implemented because the macro-quad behaviour is already smooth across the validation set.

The empirical default settings have been deliberately conservative since 2026-05-14: the rigid-TS PCOMP $\mathbf{C}_s$ scale is 2.5, the MacNeal RBF zb-scale is 0.65, the localisation filter and cluster

guard are off unless explicitly requested, and the bilinear MITC4 kernel with ϕ_2 anti-locking remains the default for all flat shells. The recommended manual override for the SOL 105 airframe ladder is `LITERAL`-rigid TS plus `Cs=2`, which brings mean/max parity to roughly 3.26% / 7.73% against Reference (against the unforced default of 6.38% / 17.81%).

15 Conclusions

The JFEM shell layer is a compact but unusually feature-rich code base for a research solver. The default kernel (bilinear MITC4 with ϕ_2 anti-locking) is what runs in production, but the dispatcher is built so that DKQ, DKMQ, Adini, MIN4, Hu–Washizu, hybrid-stress, MacNeal-RBF and MITC4-3D degenerate-shell kernels can all be selected on a per-element basis through a small set of named gates. The same gates control which membrane field, which curvature treatment and which $\mathbf{K}_G$ stress-field mode is used. This document is the per-line companion to that machinery: together with the previous architectural manuscript it should make it possible to read any line of the shell-related source and place it on the right rung of the kernel/gate/material ladder.

The two open research directions are: (i) promoting the MITC4-3D path to the default for curved laminated shells, which is the structural change expected to close the remaining largest parity gap against Reference; and (ii) replacing the higher-order `CTRIA6`/`CQUAD8` subdivision with dedicated quadratic kernels. Neither is required for the present verified capabilities, and both are tracked through the same solver-diagnostics and adjoint-registry infrastructure that the rest of the shell layer already uses.

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A Cross-reference of shell-related source files

B Minimal smoke tests for the shell paths

The shell-related smoke tests that exercise the kernels in this document include public decks under `validation/cases/` and independent reference comparisons. In summary:

- `validation/cases/macneal_harder/*.bdf` — public curved-shell and distorted-shell benchmarks for the linear shell path.
- `validation/cases/classical/timoshenko_plate_buckling.bdf` and `validation/cases/classical/cylinder_axial_buckling.bdf` — public classical buckling references.
- `validation/cases/crm/wingbox_modal_with_props.bdf` — publishable CRM-derived shell modal coverage.
- Independent SOL 105 comparisons and per-element probes exercise reference parity, `PCOMP` curvature effects, and static-preload geometric-stiffness diagnostics.

Table 8: MacNeal RBF shear and MITC4-3D-aspect gates.

Variable	Default	Effect
JFEM_Q4_MACNEAL_WARP_TOL	10^{-4}	Maximum warp ratio for MacNeal eligibility.
JFEM_Q4_MACNEAL_ASPECT_MAX	10^{30}	Aspect-ratio cap.
JFEM_Q4_MACNEAL_BENDING_SCALE	1.0	MacNeal bending multiplier.
JFEM_Q4_MACNEAL_BENDING_ISOLATED_SCALE	matches	Multiplier for isolated elements (valence 1).
JFEM_Q4_MACNEAL_CURVED_BENDING_SCALE	1.0	MacNeal bending multiplier on curved elements.
JFEM_Q4_MACNEAL_CURVED_BENDING_KAPPA_L_MIN	10^{-6}	Curvature floor for the curved-bending scale.
JFEM_Q4_MACNEAL_CURVED_BENDING_CYL_RATIO_MAX	1.0	Cylindricity cap.
JFEM_Q4_MACNEAL_PCOMP_SURFACE_KAPPA_L_MAX	10^{-4}	Maximum curvature for MacNeal PCOMP (planarity gate).
JFEM_Q4_MACNEAL_RIGID_SHEAR_FORCE	false	Force rigid transverse shear on MacNeal kernel.
JFEM_Q4_MACNEAL_BENDING_ASPECT_SCALE_*	—	Aspect-ratio interpolation series.
JFEM_Q4_MACNEAL_BENDING_ASPECT2_SCALE_*	—	Second aspect-ratio interpolation series.
JFEM_Q4_MACNEAL_EPSILON	0.04	Aspect-ratio interpolation ε.
JFEM_Q4_MACNEAL_RBF_ZB_SCALE	0.65	RBF magnitude (2026-05-14 default; LEGACY=true restores 1.0).
JFEM_Q4_MITC4_3D_ASPECT_MIN	3.0	Lower aspect-ratio bound for the aspect-gated MITC4-3D kernel.
JFEM_Q4_MITC4_3D_ASPECT_MAX	10^{30}	Upper bound.
JFEM_Q4_MITC4_3D_ASPECT_WARP_MIN/_MAX	$0/10^{30}$	Warp window.
JFEM_Q4_MITC4_3D_ASPECT_KAPPA_L_MIN/_MAX	$0/10^{30}$	Curvature window.
JFEM_Q4_MITC4_3D_ASPECT_SKEW_MIN/_MAX	—	Edge-skew window.
JFEM_Q4_MITC4_3D_ASPECT_PCOMP_ONLY	true	Restrict MITC4-3D aspect to PCOMP elements only.
JFEM_Q4_MITC4_3D_PLY_INTEGRATION	true	Use per-ply integration on PCOMP elements.
JFEM_Q4_MITC4_3D_MEMBRANE/_BENDING/_SHEAR/_DRILL_SCALE	1.0	Per-block multipliers on the MITC4-3D kernel.
JFEM_Q4_MITC4_3D_SHEAR_MODE	covariant	Shear treatment in MITC4-3D: covariant, physical, macneal/rbf.
JFEM_Q4_MITC4_3D_BENDING_ZETA_DELTA	10^{-4}	Through-thickness perturbation.

Table 9: PARAM,K6ROT and per-element drill-stiffness overrides.

Variable	Default	Effect
PARAM,K6ROT	100.0	Penalty scale on the drilling DOF; $\alpha_{\text{drill}} = (k_6/10^5)\, G\, h$.
JFEM_PARAM_K6ROT_STATIC / _EIG	inherits	Per-pass override of PARAM,K6ROT.
JFEM_PARAM_K6ROT_EIG_FLOOR	300.0	Lower bound for the eigenvalue-side K6ROT when auto-scaled.
JFEM_SHELL_SOL101_ISO_STATIC_K6ROT_OVERRIDE	—	Direct override on isotropic shells in static assembly.
JFEM_SHELL_SOL105_ISO_EIG_K6ROT_OVERRIDE	—	Direct override on isotropic shells in eigenvalue assembly.
JFEM_SHELL_SOL105_ISO_EIG_K6ROT_CYL_RATIO_MIN	—	Cylindricity floor to apply the K6ROT override.

Table 10: Static-component scaling family. Each multiplier is applied per element when the element passes the PID/EID/cluster/V2 filters.

Variable	Default	Effect
`JFEM_Q4_STATIC_COMPONENT_CM_SCALE`	1.0	Multiplier on $\mathbf{C}_m$ for matched elements.
`JFEM_Q4_STATIC_COMPONENT_CB_SCALE`	1.0	Multiplier on $\mathbf{C}_b$.
`JFEM_Q4_STATIC_COMPONENT_CS_SCALE`	1.0	Multiplier on $\mathbf{C}_s$.
`JFEM_Q4_STATIC_COMPONENT_BMB_SCALE`	1.0	Multiplier on $\mathbf{B}_{mb}$.
`JFEM_Q4_STATIC_COMPONENT_DRILL_SCALE`	1.0	Multiplier on the drill stiffness.
`JFEM_Q4_STATIC_COMPONENT_PID_FILTER`	`""`	Comma-separated `PID` list.
`JFEM_Q4_STATIC_COMPONENT_EID_FILTER`	`""`	Comma-separated EID list.
`JFEM_Q4_STATIC_COMPONENT_NEIGHBOR_PID_PREFIXES`	`""`	Cluster-guard prefixes (e.g. `WING_,FUSE_`).
`JFEM_Q4_STATIC_COMPONENT_V2_MIN/_MAX`	0.0	EIGRL V_2 window in which the scaling activates.
`JFEM_Q4_STATIC_COMPONENT_THICKNESS_MIN/_MAX`	0.0	Thickness window.
`JFEM_Q4_STATIC_COMPONENT_PCOMP_SHEAR_RATIO_MIN/_MAX`	0.0	PCOMP shear-ratio window.
`JFEM_Q4_STATIC_COMPONENT_PCOMP_D16_RATIO_MIN/_MAX`	0.0	PCOMP D_{16} window.
`JFEM_Q4_STATIC_COMPONENT_PCOMP_B_RATIO_MIN/_MAX`	0.0	PCOMP coupling-ratio window.

Table 11: PCOMP transverse-shear limit and override.

Variable	Default	Effect
`JFEM_PCOMP_RIGID_TS_DISABLE`	`false`	Restore the pre-2026-05-14 Whitney baseline ($\kappa < 1$ softening).
`JFEM_PCOMP_RIGID_TS_CS_SCALE`	2.5	Uniform $\mathbf{C}_s = s \cdot \mathbf{C}_s^{\text{Whitney}}$ multiplier when the qualifying MAT8/blank-G1Z convention is detected.
`JFEM_PCOMP_RIGID_TS_CS_FACTOR`	3.57 (when active)	Layup-aware variant: $\mathbf{C}_s = s \cdot \kappa \cdot \mathbf{C}_s^{\text{Whitney}}$.
`JFEM_PCOMP_TS_T`	5/6	Explicit override of k_s on `PCOMP`.
`JFEM_PCOMP_WHITNEY_SHEAR_ENABLED`	`true`	Toggle the per-layup Whitney–Pagano correction.

Table 12: Marguerre slope/coupling switches.

Variable	Default	Effect
`JFEM_Q4_MARGUERRE_COUPLING`	—	Enable Ibrahimbegović-style rotation-column coupling [17, 18].
`JFEM_Q4_MARGUERRE_COUPLING_SCALE`	1.0	Slope tensor scale.
`JFEM_Q4_MARGUERRE_COUPLING_CONVENTION`	`:jfem_kl` or `:handover`	Convention encoded in the trailing entry of the slope vector.
`JFEM_Q4_MARGUERRE_STATIC_COUPLING`	`false`	Allow the Marguerre coupling on the static assembly path (not just SOL 105).
`JFEM_Q4_MARGUERRE_STATIC_GEOM_NORMALS`	`true`	Use the geometric corner-normal field instead of `SNORM` when building the slope tensor.

Table 13: Geometric-stiffness gates relevant to shells.

Variable	Default	Effect
`JFEM_KG_QUAD4_STRESS_FIELD_MODE`	`auto`	$\mathbf{K}_G$ stress-field source: `auto`, `planar`, `gauss`, `avg`, `covariant`.
`JFEM_KG_QUAD4_MEMBRANE_RECOVERY_MODE`	inherits	Detailed recovery mode (`:planar`, `:covariant`, `:tri_aspect`...).
`JFEM_KG_QUAD4_COVARIANT_BLEND`	1.0	Blend between covariant and planar recovery.
`JFEM_KG_QUAD4_COVARIANT_AUTO_KAPPA_L_MIN` / `_CYL_RATIO_MAX`	—	Curvature/cylindricity gates for auto-covariant.
`JFEM_KG_QUAD4_AUTO_AVG_LOAD_CLASSIFIER`	`true`	Activate the load classifier described in §9.
`JFEM_KG_QUAD4_AUTO_AVG_REQUIRE_GEOMETRY`	`false`	Require the geometry test to pass before switching to averaged stress.
`JFEM_KG_QUAD4_AUTO_AVG_REQUIRE_COMPRESSION`	`true`	Require both principal averages to be in compression.
`JFEM_KG_QUAD4_AUTO_AVG_REQUIRE_BIAXIAL_COMPRESSION`	`true`	Require biaxial compression.
`JFEM_KG_QUAD4_AUTO_AVG_LOAD_AXIS`	1 (X)	Dominance axis for the auto-load classifier.
`JFEM_KG_QUAD4_AUTO_AVG_LOAD_SIGN`	-1 (compression)	Sign of the dominant load.
`JFEM_KG_QUAD4_AUTO_AVG_LOAD_DOMINANCE_MIN`	0.9	Fraction of the total load that the dominant axis must carry.
`JFEM_KG_QUAD4_AUTO_AVG_LOAD_TRANSVERSE_RATIO_MAX`	0.1	Maximum allowed transverse-load fraction.
`JFEM_KG_QUAD4_SHEAR_AVG_OPERATOR`	`false`	Average Gauss-point shears at corners for $\mathbf{K}_G$.
`JFEM_KG_SHELL_TRANS_MODE`	`:none`	Add principal transverse-shear coupling.
`JFEM_KG_SHELL_PRINCIPAL_TRANSVERSE_FLAT_ONLY`	`true`	Restrict to flat elements.
`JFEM_KG_SHELL_PRINCIPAL_SHEAR_YY_FACTOR`	—	YY shear factor.
`JFEM_KG_SHELL_PRINCIPAL_SHEAR_XY_FACTOR`	—	XY shear factor.
`JFEM_KG_SHELL_PRINCIPAL_SHEAR_Z_FACTOR`	—	Z transverse factor.
`JFEM_KG_SHELL_ROT_GRAD_SCALE`	1.0	Rotation-gradient contribution scale.
`JFEM_KG_SHELL_ROT_GRAD_AUTO_ISO_SCALE`	—	Auto-scale on isotropic with curvature gates.
`JFEM_KG_SHELL_ROT_GRAD_AUTO_PCOMP_SCALE`	—	Auto-scale on PCOMP.
`JFEM_KG_SHELL_PCOMP_NXY_SCALE`	1.0	PCOMP N_{xy} multiplier.
`JFEM_KG_USE_COMPATIBLE_MEMBRANE_STRESS`	inherits	Reuse the static-K compatible membrane field in $\mathbf{K}_G$.
`JFEM_KG_MEMBRANE_INCOMP`	inherits	Include Wilson bubble contribution in the $\mathbf{K}_G$ stress field.

Table 14: Files and line ranges referenced in this document.

File	Lines	Role
src/FEMKernels.jl	1–40	Module preamble, PHI2_ALPHA, MITC4PLUS_ALLOW_CURVED.
src/FEMKernels.jl	327–900	MITC4-3D shear-tying, displacement and resultant operators.
src/FEMKernels.jl	1326–1506	Bilinear MITC4 kernel stiffness_quad4_default_generic.
src/FEMKernels.jl	1746–1907	Membrane enhanced-strain.
src/FEMKernels.jl	2058–2327	Hybrid-stress (membrane, bending, combined).
src/FEMKernels.jl	2330–2487	MITC shear-drill matrices.
src/FEMKernels.jl	2643–2947	Hu–Washizu and full-feature variants.
src/FEMKernels.jl	2948–3777	Main stiffness_quad4_matrices orchestrator.
src/FEMKernels.jl	3854–4039	DKQ edge relation.
src/FEMKernels.jl	4057–4261	DKMQ edge relation and exact side shear.
src/FEMKernels.jl	4263–4449	MIN4 (Tessler–Hughes).
src/FEMKernels.jl	4451–4759	MacNeal RBF and warp correction.
src/FEMKernels.jl	4894–5070	Adini rectangular plate.
src/FEMKernels.jl	5072–5577	$\mathbf{K}_G$ plate variants (Adini, DKQ, DKMQ).
src/FEMKernels.jl	5580–5654	Centre transverse-shear recovery.
src/FEMKernels.jl	5656–5842	Quad stress–strain recovery.
src/FEMKernels.jl	5994–6764	Membrane force-field recovery family.
src/FEMKernels.jl	6782–7092	CTRIA3 macro-quad triangle.
src/FEMKernels.jl	7154–7855	Quad and triangle $\mathbf{K}_G$ kernels.
src/FEMKernels.jl	7857–7935	Shell consistent mass.
src/solver/assembly.jl	1–260	Helper definitions, kg_quad4_* classifier.
src/solver/assembly.jl	1900–2200	Gate readers and per-element configuration.
src/solver/assembly.jl	2224–2273	CTRIA6/CQUAD8 sub-element expansion.
src/solver/assembly.jl	2336–2477	Per-element material assembly.
src/solver/assembly.jl	2608–3803	CQUAD4 dispatcher and assembly loop.
src/solver/assembly.jl	3813–3910	CTRIA3 assembly loop.
src/solver/assembly.jl	4333–4700	assemble_geometric_stiffness for shells.
src/parsing/extract_properties.jl	1–380	PSHELL, PSHEAR, PCOMP, PBAR, PBARL, PROD, PSOLID.
src/ModelBuilder.jl	460–680	PCOMP CLT reduction, MAT8 rigid-TS limit, equivalent MAT1.
src/solver/stress_recovery.jl	163–430	Shell stress recovery.